THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET A

Chapter 8: **Introduction to Trigonometry** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{4}{5}$	Sides 4, 3, 5 ⇒ $\sin A = \frac{4}{5}$
2	**(a)** $\sqrt{3}$	$\frac{2/\sqrt{3}}{2/3} = \sqrt{3} = \tan 60^\circ$
3	**(c)** 3	$1 + 1 + 1$
4	**(c)** 60°	$\cos A = \frac{1}{2}$
5	**(b)** Incorrect; $\sin\theta \le 1$ since hypotenuse is the longest side	Perpendicular < hypotenuse
6	**(b)** 1	$\sec^2 A \cos^2 A = 1$
7	**(b)** Student B is correct	$\sin 2A \neq 2\sin A$ in general
8	**(a)** $\frac{5}{12}$	Sides 12, 5, 13
9	**(d)** 0	$\frac{1-1}{1+1} = 0$
10	**(c)** $\tan^2\theta$	Identity $1 + \tan^2\theta = \sec^2\theta$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$= \frac{1-\sin^2\theta}{1-\cos^2\theta} = \frac{\cos^2\theta}{\sin^2\theta} = \cot^2\theta = \frac{49}{64}$	1+1
12	$\frac{\frac{5}{4}+\frac{16}{3}-1}{1} = \frac{15+64-12}{12} = \frac{67}{12}$	1+1
13	$A + B = 60^\circ,\ A - B = 30^\circ$ ⇒ $A = 45^\circ,\ B = 15^\circ$	1+1
14	LHS $= \sin 60^\circ = \frac{\sqrt{3}}{2} \approx 0.87$; RHS $= \frac{1}{2} + \frac{1}{2} = 1$. Not equal ⇒ claim is **false**	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AC = 25$. (i) $\sin A = \frac{7}{25},\ \cos A = \frac{24}{25}$ (ii) $\sin C = \frac{24}{25},\ \cos C = \frac{7}{25}$	1+1+1
16	LHS $= \frac{(1-\cos\theta)^2}{\sin^2\theta} = \frac{(1-\cos\theta)^2}{(1-\cos\theta)(1+\cos\theta)} = \frac{1-\cos\theta}{1+\cos\theta}$	1+1+1
17	LHS $= \frac{\cos^2 A + 1 + 2\sin A + \sin^2 A}{(1+\sin A)\cos A} = \frac{2(1+\sin A)}{(1+\sin A)\cos A} = 2\sec A$	1+1+1
	OR: Multiply inside by $\frac{1+\sin A}{1+\sin A}$: $\sqrt{\frac{(1+\sin A)^2}{\cos^2 A}} = \frac{1+\sin A}{\cos A} = \sec A + \tan A$	1+1+1
18	$\tan A = \frac{3}{4}$: LHS $= \frac{7/16}{25/16} = \frac{7}{25}$; $\cos A = \frac{4}{5},\ \sin A = \frac{3}{5}$: RHS $= \frac{16-9}{25} = \frac{7}{25}$ ✓ Yes	1+1+1
	OR: $PR^2 - QR^2 = 25 \Rightarrow PR - QR = 1$ ⇒ $PR = 13,\ QR = 12$. $\sin P = \frac{12}{13},\ \cos P = \frac{5}{13},\ \tan P = \frac{12}{5}$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Write in sin, cos: $\frac{\sin^2\theta}{\cos\theta(\sin\theta-\cos\theta)} - \frac{\cos^2\theta}{\sin\theta(\sin\theta-\cos\theta)} = \frac{\sin^3\theta-\cos^3\theta}{\sin\theta\cos\theta(\sin\theta-\cos\theta)} = \frac{1+\sin\theta\cos\theta}{\sin\theta\cos\theta} = 1 + \sec\theta\csc\theta$	2+2+1
	OR: LHS $= \sin^2 A + 2 + \csc^2 A + \cos^2 A + 2 + \sec^2 A = 5 + (1 + \cot^2 A) + (1 + \tan^2 A) = 7 + \tan^2 A + \cot^2 A$	2+2+1
20	(i) $\sqrt{144 + 25} = 13$ m (ii) $\frac{5}{13}, \frac{12}{13}, \frac{5}{12}$ (iii) $\frac{25+144}{169} = 1$ ✓ (iv) False: $\tan\theta = \frac{5}{12} < 1$ (rise < base)	1+2+1+1
	OR: $\sec^2\theta - \tan^2\theta = 1 \Rightarrow \sec\theta - \tan\theta = \frac{1}{p}$. So $\sec\theta = \frac{p^2+1}{2p}$, $\tan\theta = \frac{p^2-1}{2p}$; $\sin\theta = \frac{\tan\theta}{\sec\theta} = \frac{p^2-1}{p^2+1}$	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME — SET B

Chapter 8: **Introduction to Trigonometry** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $\frac{12}{13}$	$\sqrt{1-\frac{25}{169}}$
2	**(b)** 1	$\frac{1}{2}+\frac{1}{2}$
3	**(b)** $\cos 60^\circ$	$\frac{2/3}{4/3}=\frac{1}{2}$
4	**(b)** 1	$A=45^\circ$: $2\cdot\frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}$
5	**(b)** Incorrect; $\cos\theta$ decreases from 1 to 0	$\cos 0^\circ=1,\ \cos 90^\circ=0$
6	**(b)** 1	$\frac{(1-\sin A)(1+\sin A)}{\cos^2 A}=1$
7	**(a)** Student A is correct	$\frac{1}{\sqrt{3}}=\tan 30^\circ$
8	**(a)** $\frac{5}{12}$	$BC=12$; $\tan C=\frac{AB}{BC}$
9	**(b)** 9	$9(\sec^2 A-\tan^2 A)$
10	**(a)** 30°	$\sin\theta=\frac{1}{2}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Base 12, hyp 13, perp 5: $\sin=\frac{5}{13},\cos=\frac{12}{13},\tan=\frac{5}{12},\cot=\frac{12}{5},\csc=\frac{13}{5}$	2
12	$\frac{\frac{3}{2}-\frac{2}{\sqrt{3}}}{\frac{3}{2}+\frac{2}{\sqrt{3}}}=\frac{3\sqrt{3}-4}{3\sqrt{3}+4}=\frac{43-24\sqrt{3}}{11}$	1+1
13	$A+B=90^\circ,\ A-B=30^\circ\Rightarrow A=60^\circ,\ B=30^\circ$	1+1
14	In right $\triangle ABC$, $\sin^2 A+\cos^2 A=\frac{BC^2+AB^2}{AC^2}=1$ by Pythagoras — true for every acute angle (e.g. $\theta=30^\circ$: $\frac{1}{4}+\frac{3}{4}=1$).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\sin\theta=\frac{3}{5},\ \cos\theta=\frac{4}{5}$: $\frac{\frac{12}{5}-\frac{4}{5}+1}{\frac{12}{5}+\frac{4}{5}-1}=\frac{13/5}{11/5}=\frac{13}{11}$	1+1+1
16	LHS $=\cos A+1$; RHS $=\frac{(1-\cos A)(1+\cos A)}{1-\cos A}=1+\cos A$ ✓	1+1+1
17	Divide by $\sin A$: $\frac{\cot A+\csc A-1}{\cot A-\csc A+1}$; write $1=\csc^2 A-\cot^2 A$ in the numerator: $\frac{(\csc A+\cot A)(1-\csc A+\cot A)}{\cot A-\csc A+1}=\csc A+\cot A$	1+1+1
	OR: $\frac{\sec^2 A}{\csc^2 A}=\tan^2 A$; $\frac{1-\tan A}{1-\frac{1}{\tan A}}=\frac{(1-\tan A)\tan A}{\tan A-1}=-\tan A$, squared $=\tan^2 A$	1½+1½
18	$A=30^\circ,\ C=60^\circ$: $\frac{1}{2}\cdot\frac{1}{2}+\frac{\sqrt{3}}{2}\cdot\frac{\sqrt{3}}{2}=1$	1+1+1
	OR: $\sin A=\frac{1}{\sqrt{1+\cot^2 A}}$, $\sec A=\frac{\sqrt{1+\cot^2 A}}{\cot A}$, $\tan A=\frac{1}{\cot A}$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	LHS $=\frac{(\sec\theta-1)+(\sec\theta+1)}{\sqrt{\sec^2\theta-1}}=\frac{2\sec\theta}{\tan\theta}=\frac{2}{\cos\theta}\cdot\frac{\cos\theta}{\sin\theta}=2\csc\theta$	2+2+1
	OR: LHS $=\frac{\cos^2 A}{\sin A}\cdot\frac{\sin^2 A}{\cos A}=\sin A\cos A$; RHS $=\frac{1}{\frac{\sin^2 A+\cos^2 A}{\sin A\cos A}}=\sin A\cos A$ ✓	2+2+1
20	(i) 10 cm (ii) $\sin P=\frac{QR}{PR}=\frac{3}{5}$, $\cos R=\frac{QR}{PR}=\frac{3}{5}$ (iii) Both equal $\frac{QR}{PR}$; P and R are complementary (iv) $\frac{6}{8}\times\frac{8}{6}=1$	1+2+1+1
	OR: $m^2-n^2=4\tan\theta\sin\theta$; $mn=\tan^2\theta-\sin^2\theta=\sin^2\theta\left(\frac{1}{\cos^2\theta}-1\right)=\sin^2\theta\tan^2\theta\Rightarrow 4\sqrt{mn}=4\sin\theta\tan\theta$ ✓	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.